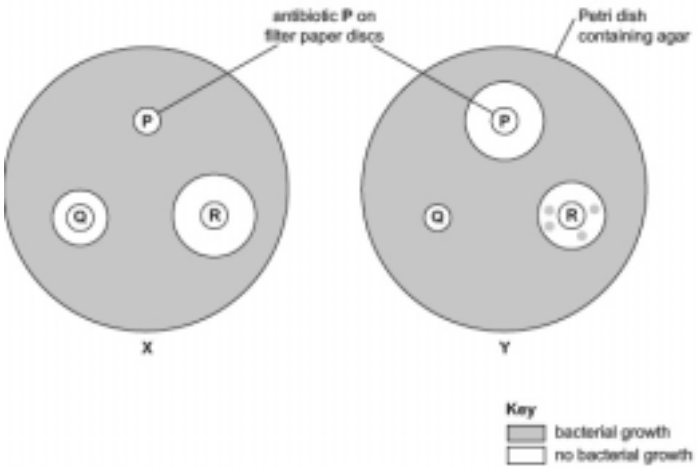


4(a)(i)	X – R <i>and</i> Y – P ;	1
4(a)(ii)	any two from: mp1 = resistance with ref. to P / Q / R Y, is resistant to Q / has no resistance to P ; A Y has some resistance to (antibiotic) R mp2 = antibiotics used at different concentrations (antibiotic) R may, have different concentration / be less effective, compared with P ; mp3 = reason for resistance ref. to gene(s) for resistance (on plasmids) or Y has, cell wall / cell membrane, that prevents entry of antibiotic Q or Y has enzyme that breaks down, Q / R ; mp4 = action of antibiotics <i>idea that</i> antibiotics have, different / specific, target(s) / AW ; A any example of process inhibited by antibiotic, e.g. cell wall synthesis transcription translation DNA replication cell surface membrane function synthesis of folic acid	2

Question	Answer	Marks
4(a)(ii)	mp5 = AVP ; e.g. <i>idea that P</i> may be bacteriocidal / (antibiotic) R is bacteriostatic e.g. <i>idea that</i> gene for resistance to antibiotic R passed by, vertical / horizontal, transmission  The diagram shows two Petri dishes, X and Y, containing agar. Each dish has three filter paper discs labeled P, Q, and R. In dish X, disc P shows a large white zone of inhibition, while discs Q and R show no inhibition. In dish Y, disc P shows a large white zone of inhibition, disc Q shows a small white zone of inhibition, and disc R shows no inhibition. A key indicates that grey shading represents bacterial growth and white represents no bacterial growth.	

Question	Answer				Marks
4(b)	any three from: to award the MP1 to MP5 you must be able to see a difference between the two stated clearly				3
		<i>vaccines</i>		<i>antibiotics</i>	
	1	(generally) are preventative / are not a treatment / are not a cure or use before a person, has an infection / is ill	v	are (generally) not preventative / are a treatment / are a cure or used when a person, has an infection / is ill ;	
	2	effective against bacteria and viruses (in context of different vaccines)	v	not effective against viruses / only effect against bacteria ;	
	3	<i>idea of</i> indirect effect on pathogens	v	<i>idea of</i> direct effect on pathogens ; A example of effect	
	4	not given as a course / give once or a few times (or with boosters)	v	given as a course / over many days / AW ;	
	5	specific for particular, pathogen	v	(most) antibiotics act on a range of pathogens ;	
	6	stimulate an <u>immune response</u>	or	do not stimulate an <u>immune response</u> ;	
	7	detail e.g. stimulate, (B- / T-) lymphocytes / production of antibodies A stimulates production of memory cells	or	do not stimulate, (B- / T-) lymphocytes / production of antibodies ; A no stimulation of memory cells	
	8	provide long-term, protection / immunity	or	do not provide long-term, protection / immunity ;	
	9	does not lead to resistance	or	may lead to resistance ;	
	10	time delay before being effective / AW	or	have effect, faster / sooner / AW ;	